NutriVision: Image-Based Food Recognition and Nutritional Estimation Using Deep Learning

Project 2 – Level 1 Summary Report

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I. Project Overview

NutriVision is a computer vision pipeline that classifies food from photographs and estimates nutritional content using the USDA FoodData Central database. It is trained on the Food-101 dataset [1], which has 101,000 images spanning 101 food categories. The end goal is a tool where a user can photograph their meal and get back an automatic breakdown of calories, protein, carbs, and fat. This report covers Level 1 of the project, where I trained and compared two deep learning models on the food classification task.

II. Level 1 Goals

1. **Dataset preparation:** Download and preprocess Food-101 (101 classes, 101K images) with augmentation (random crops, flips, rotation, color jitter) and ImageNet normalization.
2. **ResNet-50 baseline:** Train a pre-trained ResNet-50 using two-phase transfer learning. Phase 1 freezes the backbone and only trains the classification head. Phase 2 unfreezes everything and fine-tunes with differential learning rates.
3. **Architecture comparison:** Train MobileNetV3-Large under the same conditions to see how a lighter model compares in accuracy vs. speed.
4. **Evaluation:** Report Top-1 and Top-5 accuracy, per-class precision/recall/F1, a confusion matrix, and an analysis of which food pairs the model confuses the most.

Success criterion: $\geq$75% Top-1 accuracy on the Food-101 test set.

III. Results

A. Model Comparison

Table 1: Test Set Performance on Food-101

Metric	ResNet-50	MobileNetV3
Top-1 Accuracy (%)	**82.95**	76.25
Top-5 Accuracy (%)	**96.18**	93.43
Parameters (M)	23.7	**4.3**
Inference (ms/image)	5.6	**4.6**

ResNet-50 hits **82.95% Top-1 accuracy**, which clears the 75% target by a good margin. With 96.18% Top-5 accuracy, the correct food is almost always somewhere in the model's top five guesses, which will be useful for the nutrition pipeline in Level 2 (where a user could pick from the top candidates). MobileNetV3 scores 6.7% lower in accuracy but is 5.5$\times$ smaller and 18% faster at inference, so it could work well for a mobile app where speed matters more.

B. Training Dynamics

I used a two-phase training strategy for both models. In Phase 1 (5 epochs), only the classification head was trained while the backbone stayed frozen (Adam, lr $= 10^{-3}$). In Phase 2 (up to 15 epochs), the full network was fine-tuned with a lower learning rate for the backbone (10^{-5}) and a higher one for the head (10^{-4}). ResNet-50 reached 58.9% validation accuracy after Phase 1, and Phase 2 pushed it up to 77.9% as the backbone learned food-specific features. MobileNetV3 overfit more noticeably, with training accuracy climbing to around 90% while validation stayed near 70%. Figure 1 shows both models side by side.

C. Confusion Analysis

Table 2: Best and Worst Classes by F1-Score (ResNet-50)

Class	Prec.	Rec.	F1
Best Performing			
Edamame	.988	.996	.992
Macarons	.976	.968	.972
Oysters	.943	.924	.933
Spagh. Carbonara	.943	.920	.931
Seaweed Salad	.939	.920	.929
Worst Performing			
Steak	.585	.564	.574
Pork Chop	.672	.532	.594
Filet Mignon	.589	.712	.645
Choc. Mousse	.648	.664	.656
Foie Gras	.675	.640	.657

The model does best on foods that look distinct: edamame (bright green pods), macarons (colorful discs), oysters (shells on ice). The hardest classes are foods that genuinely look alike. Table 3 breaks down the worst confusion pairs. Three patterns stand out: **(1) Foods that look nearly identical on a plate**, like steak, filet mignon, and pork chop (141 total errors between them), where the dif-

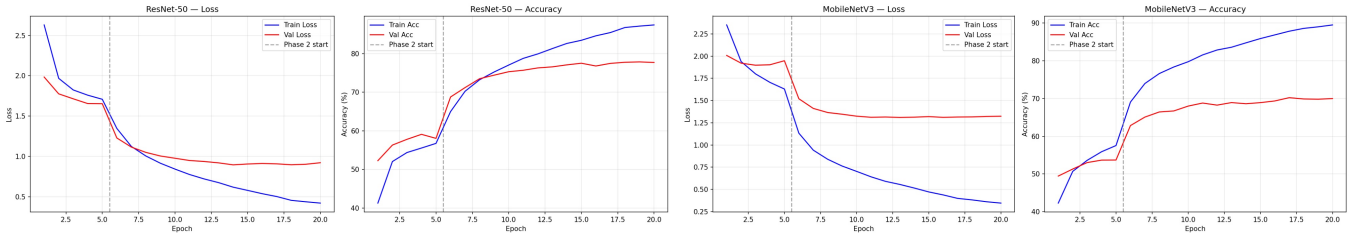

Figure 1: Training curves for ResNet-50 (left) and MobileNetV3 (right). The dashed lines mark the transition from Phase 1 to Phase 2. ResNet-50 has a tighter gap between train and val accuracy, while MobileNetV3 overfits more due to its smaller capacity.

ference comes down to thickness and marbling. **(2) Same ingredient prepared differently**, like tuna vs. beef tartare (51 errors) and chocolate cake vs. mousse (45 errors), which mainly differ in texture. **(3) Similar shapes with different fillings**, like ice cream vs. frozen yogurt (17 errors) and fried calamari vs. onion rings (13 errors).

Table 3: Top 8 Most Confused Food Pairs (ResNet-50)

True Class	Predicted As	Count
Steak	Filet Mignon	50
Tuna Tartare	Beef Tartare	33
Pork Chop	Filet Mignon	30
Filet Mignon	Steak	29
Choc. Cake	Choc. Mousse	23
Choc. Mousse	Choc. Cake	22
Apple Pie	Bread Pudding	21
Pork Chop	Steak	19

Figure 2 shows these patterns more clearly in the confusion matrix for the 15 hardest classes, and Figure 3 has some example misclassifications.

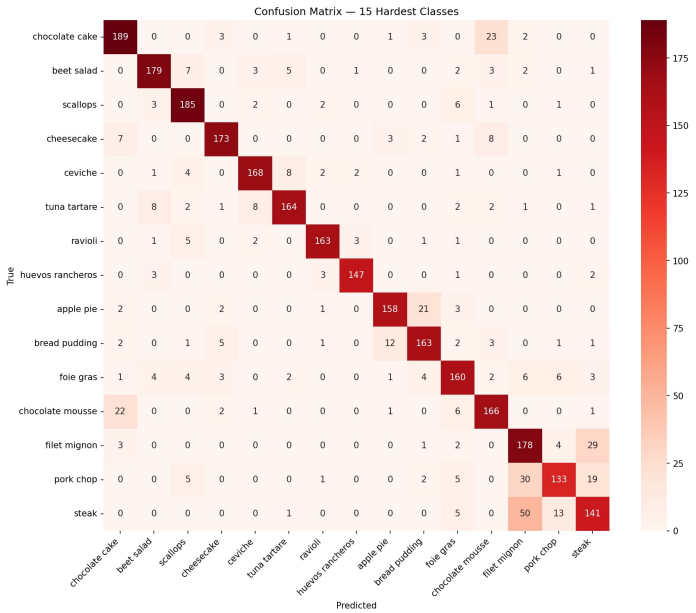

Figure 2: Confusion matrix for the 15 hardest classes. The steak/filet mignon/pork chop cluster and chocolate cake/mousse pair show the strongest two-way confusion.

Figure 3: Example misclassifications: pork chop predicted as steak, fried calamari as onion rings, nachos as apple pie. Most errors are between foods that look visually similar.

IV. Next Steps

All Level 1 goals are met. For Level 2, I plan to: (1) map all 101 food classes to the USDA FoodData Central database so the model can output nutritional estimates, (2) try cosine annealing, mixup, and label smoothing to see if accuracy improves further, (3) explore multi-food classification for plates with more than one item, and (4) put together a working demo of the full image-to-nutrition pipeline. For Level 3, I want to build a web interface (Gradio or Streamlit), add Grad-CAM visualizations to show what the model is looking at, include a text input option so users can refine results (e.g., "half portion" or "fried"), and add daily meal logging.

References

[1] L. Bossard, M. Guillaumin, and L. Van Gool, "Food-101 – Mining Discriminative Components with Random Forests," in *ECCV*, 2014.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *IEEE CVPR*, 2016, pp. 770–778.

[3] A. Howard et al., "Searching for MobileNetV3," in *IEEE/CVF ICCV*, 2019, pp. 1314–1324.